$$\begin{aligned}
&C_\phi^{(11)\ (22)\ (21)} \to \frac{1}{16\pi^2}\\
&\Big(\tfrac{3}{2} m_1 \tilde{\mu} g_1^6 \mathrm{LF}_{3,3,-2}[m_1, \tilde{\mu}] + \tfrac{1}{2} \tilde{\mu} g_1^6 m_1^3 \mathrm{LF}_{3,3,-1}[m_1, \tilde{\mu}] + \tfrac{1}{3} m_2 \tilde{\mu} g_2^4 (-g_1^2 + g_2^2) \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}] +\\
&\tfrac{1}{6} m_2 \tilde{\mu} g_2^4 (g_1^2 - g_2^2) \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] + \tfrac{11}{2} m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{3,3,-2}[m_2, \tilde{\mu}] +\\
&\tfrac{1}{2} \tilde{\mu} g_2^6 m_2^3 \mathrm{LF}_{3,3,-1}[m_2, \tilde{\mu}] - \tfrac{1}{12} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d}^{pr} (g_1^2 + 3 g_2^2) \mathrm{LF}_{2,2,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -\\
&\tfrac{1}{6} \tilde{\mu} g_1^4 y_d^{pr} \overline{a_d}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] + \tfrac{1}{4} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e}^{pr} (g_1^2 - g_2^2) \mathrm{LF}_{2,2,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] -\\
&\tfrac{1}{2} \tilde{\mu} g_1^4 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \tfrac{1}{2} \tilde{\mu} g_1^2 g_2^2 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -\\
&\tfrac{1}{8} \tilde{\mu} y_e^{pr} \overline{a_e}^{pr} (3 g_1^4 + 7 g_1^2 g_2^2 - 4 g_2^4) \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +\\
&\tfrac{1}{12} \tilde{\mu} y_e^{pr} \overline{a_e}^{pr} (3 g_1^4 + 8 g_1^2 g_2^2 - 5 g_2^4) \mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +\\
&\tfrac{1}{6} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d}^{pr} (2 g_1^2 + 3 g_2^2) \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \tfrac{3}{8} \tilde{\mu} y_d^{pr} \overline{a_d}^{pr} (3 g_1^4 + 7 g_1^2 g_2^2 - 4 g_2^4)\\
&\quad \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \tfrac{1}{4} \tilde{\mu} y_d^{pr} \overline{a_d}^{pr} (3 g_1^4 + 8 g_1^2 g_2^2 - 5 g_2^4) \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +\\
&\tfrac{1}{12} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} (2 g_1^4 - 9 g_1^2 g_2^2 + 3 g_2^4) \mathrm{LF}_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -\\
&\tfrac{1}{24} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} (g_1^4 - 6 g_1^2 g_2^2 - 9 g_2^4) \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \tfrac{1}{8} \tilde{\mu} g_2^2 y_u^{pr} \overline{a_u}^{pr} (g_1^2 - g_2^2)\\
&\quad \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \tfrac{1}{24} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} (7 g_1^4 - 24 g_1^2 g_2^2 + 15 g_2^4) \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +\\
&\tfrac{1}{4} \tilde{\mu} g_2^2 y_u^{pr} \overline{a_u}^{pr} (g_1^2 - g_2^2) \mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \tfrac{3}{8} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} (3 g_1^4 + 8 g_1^2 g_2^2 - 5 g_2^4)\\
&\quad \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \tfrac{1}{4} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} (3 g_1^4 + 8 g_1^2 g_2^2 - 5 g_2^4) \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +\\
&\tfrac{1}{8} m_1 \tilde{\mu} g_1^2 (g_1^4 + 2 g_1^2 g_2^2 - g_2^4) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_1] + \tfrac{3}{4} m_1 \tilde{\mu} g_1^4 (g_1^2 - g_2^2) \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] -\\
&\tfrac{1}{8} m_1 \tilde{\mu} g_1^2 (3 g_1^4 + 7 g_1^2 g_2^2 - 4 g_2^4) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] + \tfrac{1}{2} m_1 \tilde{\mu} g_1^4 (-g_1^2 + g_2^2) \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] +\\
&\tfrac{1}{12} m_1 \tilde{\mu} g_1^2 (3 g_1^4 + 8 g_1^2 g_2^2 - 5 g_2^4) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] +\\
&\tfrac{1}{24} m_2 \tilde{\mu} g_2^2 (9 g_1^4 + 26 g_1^2 g_2^2 - 17 g_2^4) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] + \tfrac{31}{12} m_2 \tilde{\mu} g_2^4 (g_1^2 - g_2^2) \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] +\\
&\tfrac{1}{8} m_2 \tilde{\mu} g_2^2 (-9 g_1^4 - 25 g_1^2 g_2^2 + 16 g_2^4) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] +\\
&2 m_2 \tilde{\mu} g_2^4 (-g_1^2 + g_2^2) \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \tfrac{1}{4} m_2 \tilde{\mu} g_2^2 (3 g_1^4 + 8 g_1^2 g_2^2 - 5 g_2^4) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] +\\
&\tfrac{1}{4} \tilde{\mu} g_1^2 g_2^2 (g_1^2 (7 m_1 + 5 m_2) - g_2^2 (5 m_1 + 7 m_2)) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] -\\
&\tilde{\mu} g_1^2 g_2^2 (g_1^2 - g_2^2) (m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] +\\
&\tfrac{3}{8} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -\\
&\tfrac{3}{8} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +\\
&\tfrac{1}{8} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] -\\
&\tfrac{1}{8} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +\\
&\tfrac{1}{4} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (-g_1^2 + g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] -\\
&\tfrac{1}{2} \tilde{\mu} g_2^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] +\\
&\tfrac{1}{4} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (-g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] +\\
&\tfrac{1}{8} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}] -\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +\\
&\tfrac{1}{4} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +\\
&\tfrac{1}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (g_1^2 + 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] +\\
&\tfrac{1}{2} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (2 g_1^2 - 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] +\\
&\tfrac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (-g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] -\\
&3 \tilde{\mu} g_2^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + \tfrac{3}{2} \tilde{\mu} g_2^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr}\\
&\quad \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] - \tfrac{3}{2} \tilde{\mu} g_2^2 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{u}}^{r}] +\\
&\tfrac{3}{8} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr} (g_1^2 + g_2^2) (\overline{y_d}^{st} y_d^{pt} - \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}] +\\
&\tfrac{1}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (3 \overline{y_d}^{st} y_d^{pt} (-g_1^2 + g_2^2) + \overline{y_u}^{st} y_u^{pt} (g_1^2 - 9 g_2^2)) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] +\\
&\tfrac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - g_2^2) (\overline{y_d}^{st} y_d^{pt} - \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] -\\
&\tfrac{3}{2} \tilde{\mu} g_2^2 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}] +\\
&\tfrac{3}{16} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (\overline{y_d}^{st} y_d^{pt} (g_1^2 - 3 g_2^2) + \overline{y_u}^{st} y_u^{pt} (g_1^2 + g_2^2)) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}] -\\
&\tfrac{1}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +\\
&\tfrac{3}{2} \tilde{\mu} g_2^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}] -\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +\\
&\tfrac{3}{4} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr} (\overline{y_d}^{st} y_d^{pt} (g_1^2 + g_2^2) - 2 g_2^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -\\
&\tfrac{1}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (g_1^2 + 3 g_2^2) (3 \overline{y_d}^{st} y_d^{pt} - \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +\\
&\tfrac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - g_2^2) (\overline{y_d}^{st} y_d^{pt} - \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +\\
&\tfrac{3}{16} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (\overline{y_d}^{st} y_d^{pt} (g_1^2 + g_2^2) + \overline{y_u}^{st} y_u^{pt} (-3 g_1^2 + g_2^2)) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +\\
&\tfrac{1}{4} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (3 g_2^2 \overline{y_d}^{st} y_d^{pt} + \overline{y_u}^{st} y_u^{pt} (4 g_1^2 - 3 g_2^2)) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +\\
&\tfrac{3}{4} \tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} (-\overline{y_d}^{st} y_d^{pt} + \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +\\
&\tfrac{1}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} (7 g_1^2 + 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] -\\
&\tfrac{3}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] +\\
&\tfrac{1}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +\\
&\tfrac{3}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -\\
&\tfrac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 (g_1^2 + g_2^2) \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_1, m_2] +\\
&\tfrac{1}{8} \tilde{\mu} g_1^2 g_2^2 (g_2^2 (11 m_1 - 17 m_2) + g_1^2 (11 m_1 + 7 m_2)) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^4 g_2^2 (2 m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_1, m_2] + \tfrac{1}{2} m_1 \tilde{\mu} g_1^4 g_2^2 (m_1 m_2 + 8 \tilde{\mu}^2) \mathrm{LF}_{3,2,1,-1}[\tilde{\mu}, m_1, m_2] -\\
&\tfrac{3}{4} \tilde{\mu} g_1^2 g_2^2 (g_2^2 (m_1 - 3 m_2) + g_1^2 (m_1 + m_2)) \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 g_2^2 (-2 g_1^2 (m_1 + m_2) + g_2^2 (5 m_1 + 6 m_2)) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] +\\
&\tfrac{1}{2} m_2 \tilde{\mu} g_1^2 g_2^4 (-3 m_1 m_2 + 8 \tilde{\mu}^2) \mathrm{LF}_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 y_d^{sr} \overline{a_d}^{pr} (\overline{a_d}^{st} a_d^{pt} - \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +\\
&3 \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{v}] -\\
&\tfrac{3}{8} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] +\\
&3 \tilde{\mu} \overline{y_d}^{st} y_d^{sr} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}] +\\
&3 \tilde{\mu} y_d^{sr} \overline{y_u}^{st} \overline{y_u}^{uv} y_u^{pv} y_u^{ut} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}] -\\
&\tfrac{3}{8} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 y_d^{sr} \overline{a_d}^{pr} (\overline{a_d}^{st} a_d^{pt} - \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +\\
&\tfrac{3}{8} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] +\\
&3 \tilde{\mu} \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{v}] +\\
&\tfrac{3}{8} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +\\
&3 \tilde{\mu} \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_u^{ur} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 y_e^{sr} \overline{a_e}^{pr} (\overline{a_e}^{st} a_e^{pt} - \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] -\\
&\tfrac{1}{8} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] -\\
&\tfrac{1}{8} \tilde{\mu} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +\\
&\tfrac{1}{2} \tilde{\mu} g_1^2 y_e^{sr} \overline{a_e}^{pr} (\overline{a_e}^{st} a_e^{pt} - \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] +\\
&\tfrac{1}{8} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] +\\
&\tfrac{1}{8} \tilde{\mu} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +\\
&\tfrac{1}{4} \tilde{\mu} y_e^{sr} \overline{a_e}^{pr} (-2 g_2^2 \overline{a_e}^{st} a_e^{pt} + \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt} (g_1^2 - g_2^2)) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] +\\
&\tfrac{1}{4} \tilde{\mu} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} (-g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] +\\
&\tfrac{1}{4} \tilde{\mu} y_e^{sr} \overline{a_e}^{pr} (\overline{a_e}^{st} a_e^{pt} (-g_1^2 + g_2^2) + 2 g_1^2 \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +\\
&\tfrac{1}{4} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} (-g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] - \tfrac{1}{4} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr}\\
&\quad (2 \overline{a_d}^{st} a_d^{pt} (-2 g_1^2 + 3 g_2^2) + \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} (g_1^2 + 3 g_2^2)) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] +\\
&\tfrac{3}{4} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} (-g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] +\\
&3 \tilde{\mu} g_2^2 y_u^{pt} \overline{a_d}^{pr} (-a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +\\
&\tfrac{3}{2} \tilde{\mu} g_2^2 y_u^{pt} \overline{a_d}^{pr} (a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +\\
&\tfrac{3}{2} g_2^2 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] -\\
&\tfrac{3}{8} \tilde{\mu} \overline{a_d}^{pr} (\overline{a_u}^{st} (y_d^{sr} a_u^{pt} (g_1^2 + g_2^2) + y_u^{pt} a_d^{sr} (g_1^2 - 3 g_2^2)) + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} (g_1^2 + g_2^2))\\
&\quad \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] -\\
&\tfrac{3}{8} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - 3 g_2^2) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{u}}^{r}] +\\
&3 \tilde{\mu} \overline{y_d}^{st} y_d^{pt} \overline{y_u}^{uv} y_u^{sv} y_u^{ur} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}] + \tfrac{1}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr}\\
&\quad (\overline{a_u}^{st} a_u^{pt} (g_1^2 - 9 g_2^2) + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} (5 g_1^2 - 9 g_2^2)) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] -\\
&\tfrac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - g_2^2) (\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +\\
&\tfrac{3}{2} g_2^2 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] + \tfrac{3}{16} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr}\\
&\quad (\overline{a_u}^{st} a_u^{pt} (g_1^2 + g_2^2) + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} (-3 g_1^2 + g_2^2)) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +\\
&\tfrac{3}{2} \tilde{\mu} g_2^2 y_u^{pt} \overline{a_d}^{pr} (a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] -\\
&\tfrac{3}{16} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (\overline{a_u}^{st} a_u^{pt} (g_1^2 + g_2^2) + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} (g_1^2 - 3 g_2^2)) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] +\\
&\tfrac{1}{4} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr} (\overline{a_d}^{st} a_d^{pt} (g_1^2 + 3 g_2^2) + 2 g_1^2 \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] +\\
&\tfrac{3}{4} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} (-g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] -\\
&\tfrac{3}{2} \tilde{\mu} g_2^2 y_d^{sr} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] + \tfrac{1}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr}\\
&\quad (\overline{a_u}^{st} a_u^{pt} (g_1^2 + 3 g_2^2) + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} (5 g_1^2 + 3 g_2^2)) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] -\\
&\tfrac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - g_2^2) (\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] + \tfrac{3}{16} \tilde{\mu}\\
&\quad y_u^{sr} \overline{a_u}^{pr} (\overline{a_u}^{st} a_u^{pt} (-3 g_1^2 + g_2^2) + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} (g_1^2 + g_2^2)) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] -\\
&\tfrac{3}{2} \tilde{\mu} g_2^2 y_d^{sr} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] - \tfrac{3}{16} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr}\\
&\quad (\overline{a_u}^{st} a_u^{pt} (g_1^2 - 3 g_2^2) + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} (g_1^2 + g_2^2)) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \tfrac{1}{4} \tilde{\mu} y_u^{sr}\\
&\quad \overline{a_u}^{pr} (\overline{a_u}^{st} a_u^{pt} (-4 g_1^2 + 3 g_2^2) + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} (4 g_1^2 + 3 g_2^2)) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +\\
&\tfrac{3}{4} \tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} (\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] -\\
&\tfrac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} (g_1^2 - g_2^2) (\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] -\\
&\tfrac{1}{4} \tilde{\mu} g_1^2 y_u^{sr} \overline{a_u}^{pr} (-7 \overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] -\\
&\tfrac{3}{4} \tilde{\mu} g_1^2 y_u^{sr} \overline{a_u}^{pr} (\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] +\\
&3 \tilde{\mu}^3 \overline{y_d}^{st} \overline{y_d}^{uv} \overline{a_d}^{pr} (y_d^{pt} y_d^{sv} y_d^{ur} + y_d^{pv} y_d^{sr} y_d^{ut}) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}] +\\
&3 \tilde{\mu} \overline{a_d}^{pr} (y_d^{pt} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{st} a_d^{ur} + \tilde{\mu}^2 \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pv} y_d^{sr} y_d^{ut})\\
&\quad \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}] +\\
&3 \tilde{\mu} \overline{y_u}^{uv} \overline{a_d}^{pr} (y_d^{sr} y_d^{ut} \overline{a_d}^{st} a_u^{pv} - \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} y_d^{ur} y_u^{sv}) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}] -\\
&3 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}] +\\
&3 \tilde{\mu} \overline{y_d}^{uv} y_d^{pv} y_u^{ut} \overline{a_d}^{pr} a_d^{sr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +\\
&3 \tilde{\mu} \overline{a_d}^{pr} (\overline{a_u}^{st} (\overline{y_d}^{uv} y_d^{sv} y_d^{ur} a_u^{pt} + \overline{y_u}^{uv} y_u^{pv} y_u^{ut} a_d^{sr}) - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st} \overline{y_u}^{uv} y_u^{pv} y_u^{ut})\\
&\quad \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}] +\\
&3 \tilde{\mu} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}] -\\
&3 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{v}] -\\
&3 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{v}] -\\
&3 \tilde{\mu}^3 \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_u^{ur} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}] +\\
&3 \tilde{\mu} \overline{y_d}^{uv} y_u^{sr} y_u^{ut} a_d^{pv} \overline{a_u}^{pr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +\\
&\tilde{\mu}^3 \overline{y_e}^{st} \overline{y_e}^{uv} \overline{a_e}^{pr} (y_e^{pt} y_e^{sv} y_e^{ur} + y_e^{pv} y_e^{sr} y_e^{ut}) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{u}] +\\
&\tilde{\mu}^3 \overline{y_e}^{st} \overline{y_e}^{uv} y_e^{pv} y_e^{sr} y_e^{ut} \overline{a_e}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}] +\\
&3 \tilde{\mu} \overline{y_d}^{uv} y_d^{sv} y_u^{pt} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{ur} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +\\
&3 \tilde{\mu}^3 \overline{y_u}^{st} \overline{y_u}^{uv} y_u^{pt} y_u^{sv} y_u^{ur} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{v}] +\\
&3 \tilde{\mu}^3 \overline{y_u}^{st} \overline{y_u}^{uv} \overline{a_u}^{pr} (y_u^{pt} y_u^{sv} y_u^{ur} + y_u^{pv} y_u^{sr} y_u^{ut}) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}] +\\
&3 \tilde{\mu}^3 \overline{y_d}^{uv} \overline{a_d}^{pr} \overline{a_d}^{st} (y_d^{sv} y_d^{ur} a_d^{pt} + y_d^{sr} y_d^{ut} a_d^{pv}) \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}] +\\
&3 \tilde{\mu} \overline{a_d}^{pr} (\tilde{\mu}^4 \overline{y_d}^{st} y_d^{pt} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} + \overline{a_d}^{st} (y_d^{pt} a_d^{ur} \overline{a_u}^{uv} a_u^{sv} - \tilde{\mu}^2 y_d^{sr} y_d^{ut} \overline{y_u}^{uv} a_u^{pv}))\\
&\quad \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}] -\\
&3 \tilde{\mu}^3 \overline{y_d}^{uv} y_d^{pv} y_u^{ut} \overline{a_d}^{pr} a_d^{sr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}] + 3 \tilde{\mu} \overline{a_d}^{pr}\\
&\quad \overline{a_u}^{st} (y_u^{ut} a_d^{sr} \overline{a_u}^{uv} a_u^{pv} - \tilde{\mu}^2 y_d^{ur} \overline{y_u}^{uv} y_u^{sv} a_u^{pt}) \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}] +\\
&3 \tilde{\mu}^5 \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{v}] -\\
&3 \tilde{\mu}^3 \overline{y_d}^{uv} y_u^{sr} y_u^{ut} a_d^{pv} \overline{a_u}^{pr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +\\
&\tilde{\mu}^3 \overline{y_e}^{uv} \overline{a_e}^{pr} \overline{a_e}^{st} (y_e^{sv} y_e^{ur} a_e^{pt} + y_e^{sr} y_e^{ut} a_e^{pv}) \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}] +\\
&3 \tilde{\mu}^3 \overline{y_u}^{uv} \overline{a_u}^{pr} \overline{a_u}^{st} (y_u^{sv} y_u^{ur} a_u^{pt} + y_u^{sr} y_u^{ut} a_u^{pv}) \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}]\Big)
\end{aligned}$$